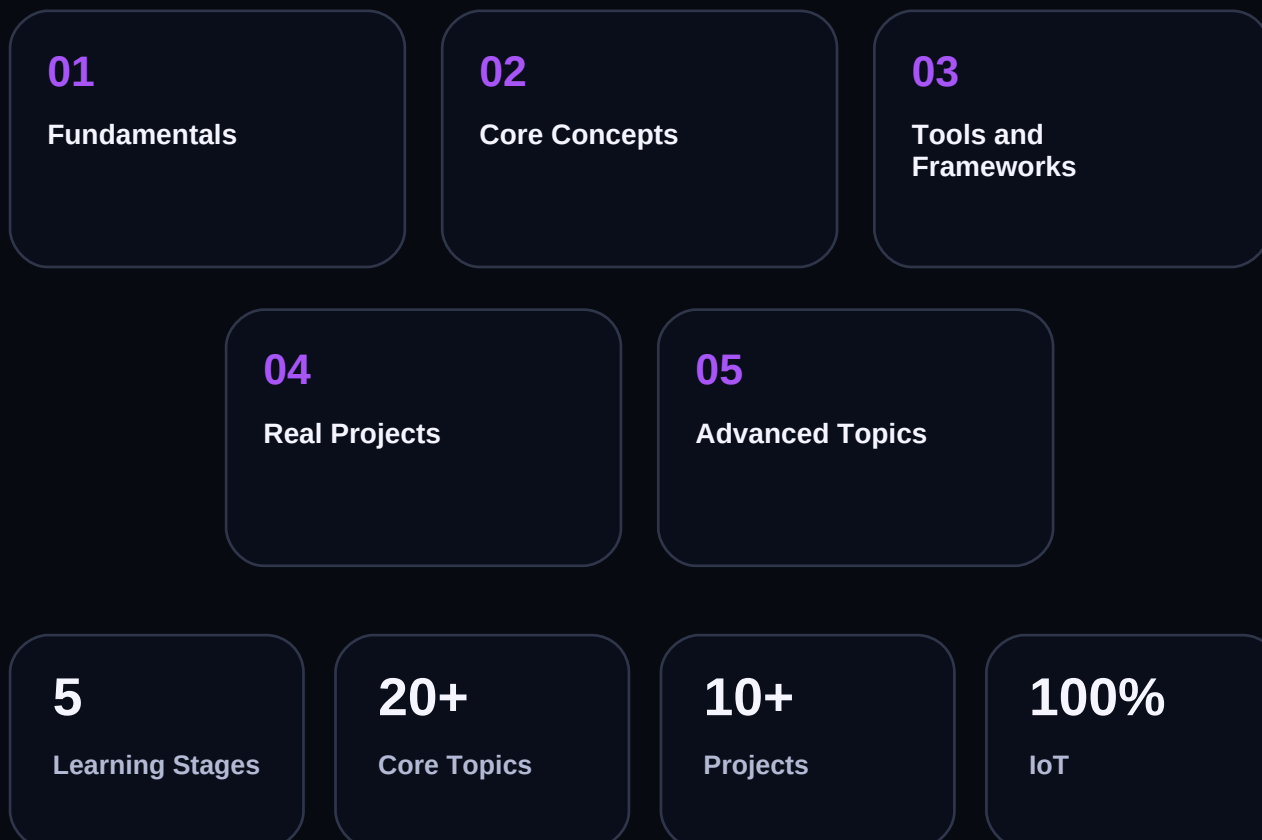
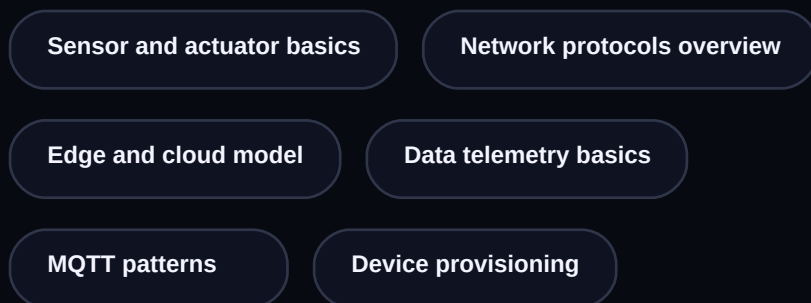


IOT DEVELOPMENT

Complete Learning Roadmap

Build end-to-end IoT systems across devices, gateways, cloud ingestion, and monitoring.



Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

Sensor and actuator basics •
Network protocols overview •
Edge and cloud model • Data
telemetry basics

02

Core Concepts

MQTT patterns • Device
provisioning • Secure
communication • Remote
command handling

03

Tools and Frameworks

IoT platform tooling •
Time-series storage • Edge
runtime tools • Rules engines

04

Real Projects

Smart environment prototype •
Predictive alerting pipeline •
Fleet dashboard • Device
provisioning workflow

05

Advanced Topics

Fleet security governance •
Offline edge resilience • Cost
and bandwidth optimization •
IoT observability strategy

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.

Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Sensor and actuator basics
- Network protocols overview
- Edge and cloud model
- Data telemetry basics

Tools & Frameworks

- Sensor and actuator basics
- Network protocols overview
- Edge and cloud model
- Data telemetry basics

Recommended Projects

- IoT Development workshop using Sensor and actuator basics
- IoT Development portfolio project with Network protocols overview

Stage 1 Milestone Complete IoT Development workshop using Sensor and actuator basics and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- MQTT patterns
- Device provisioning
- Secure communication
- Remote command handling

Tools & Frameworks

- MQTT patterns
- Device provisioning
- Secure communication
- Remote command handling

Recommended Projects

- IoT Development workshop using MQTT patterns
- IoT Development portfolio project with Device provisioning

Stage 2 Milestone Complete IoT Development workshop using MQTT patterns and document the decisions, trade-offs, and validation checks before moving to Stage 3.



Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- IoT platform tooling
- Time-series storage
- Edge runtime tools
- Rules engines

Tools & Frameworks

- IoT platform tooling
- Time-series storage
- Edge runtime tools
- Rules engines

Recommended Projects

- IoT Development workshop using IoT platform tooling
- IoT Development portfolio project with Time-series storage

Stage 3 Milestone Complete IoT Development workshop using IoT platform tooling and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- Smart environment prototype
- Predictive alerting pipeline
- Fleet dashboard
- Device provisioning workflow

Tools & Frameworks

- Smart environment prototype
- Predictive alerting pipeline
- Fleet dashboard
- Device provisioning workflow

Recommended Projects

- IoT Development workshop using Smart environment prototype
- IoT Development portfolio project with Predictive alerting pipeline

Stage 4 Milestone Complete IoT Development workshop using Smart environment prototype and document the decisions, trade-offs, and validation checks before moving to Stage 5.



Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

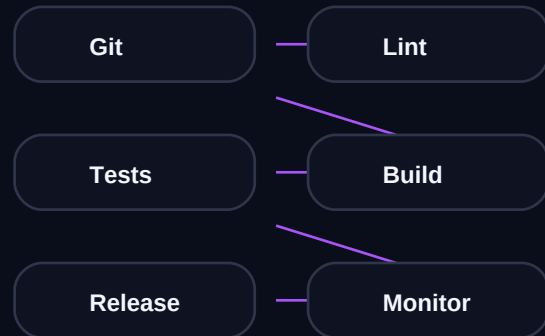
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Fleet security governance
- Offline edge resilience
- Cost and bandwidth optimization
- IoT observability strategy

CI/CD Pipeline



Stage 5 Milestone

A production-grade iot development workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

IoT Development docs and community guides

Link: <https://aws.amazon.com/iot/>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/software-architect>

PAID

Structured specialization track focused on IoT Development

Link: <https://roadmap.sh/software-architect>

FREE

IoT Development roadmap reference

Link: <https://roadmap.sh/software-architect>

FREE

Official documentation

Link: <https://aws.amazon.com/iot/>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in IoT Development?
- Which project best proves Stage 4 competency in IoT Development?
- What trade-offs appear in Stage 5 decisions for IoT Development?